

Handwritten Character Recognition

UTD Deep Dive into AI Final Project
- Kumar Samyak

MOTIVATION

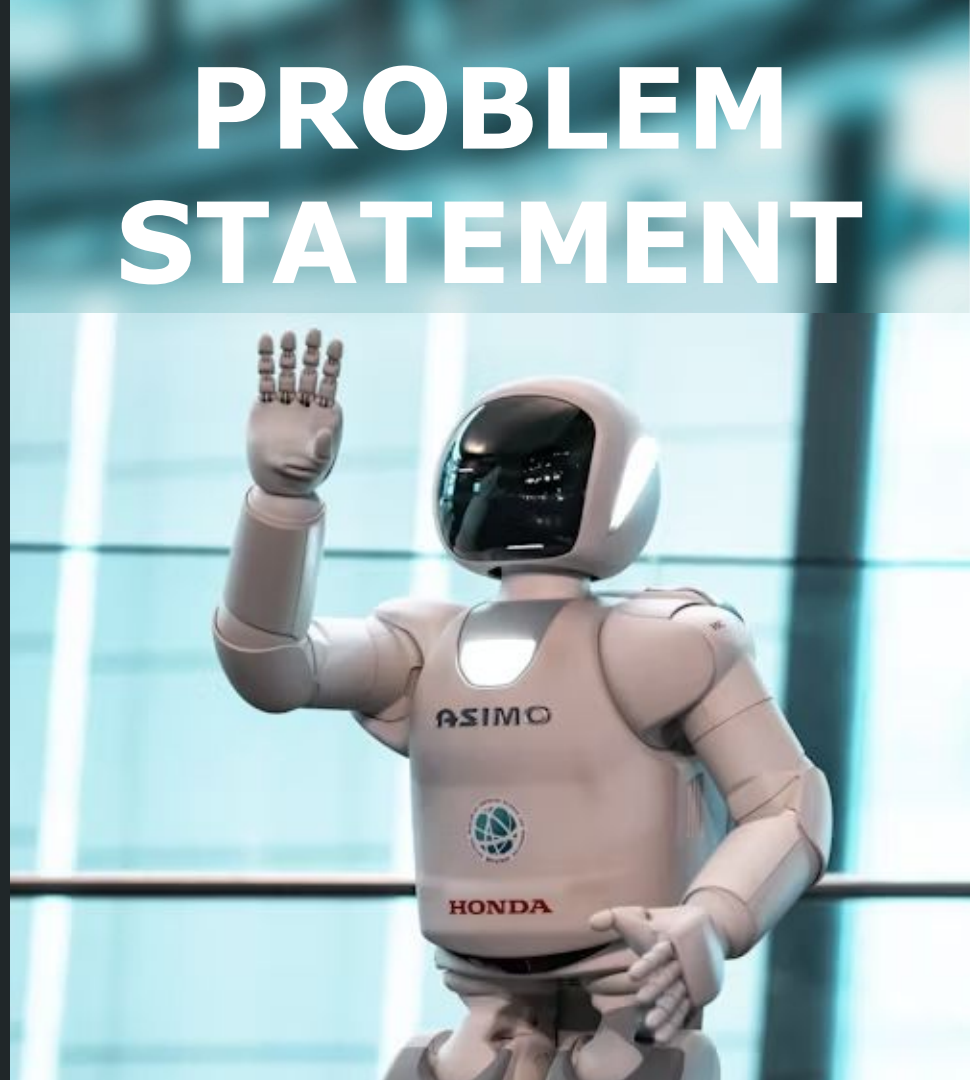


I wanted to create a model that can read handwritten text.

There are many applications for Handwritten Character Recognition (HCR) such as reading notes, translating written text, etc.

PROBLEM STATEMENT

Develop a neural network-based model that can accurately recognize and classify handwritten characters (alphabets or digits) from images.

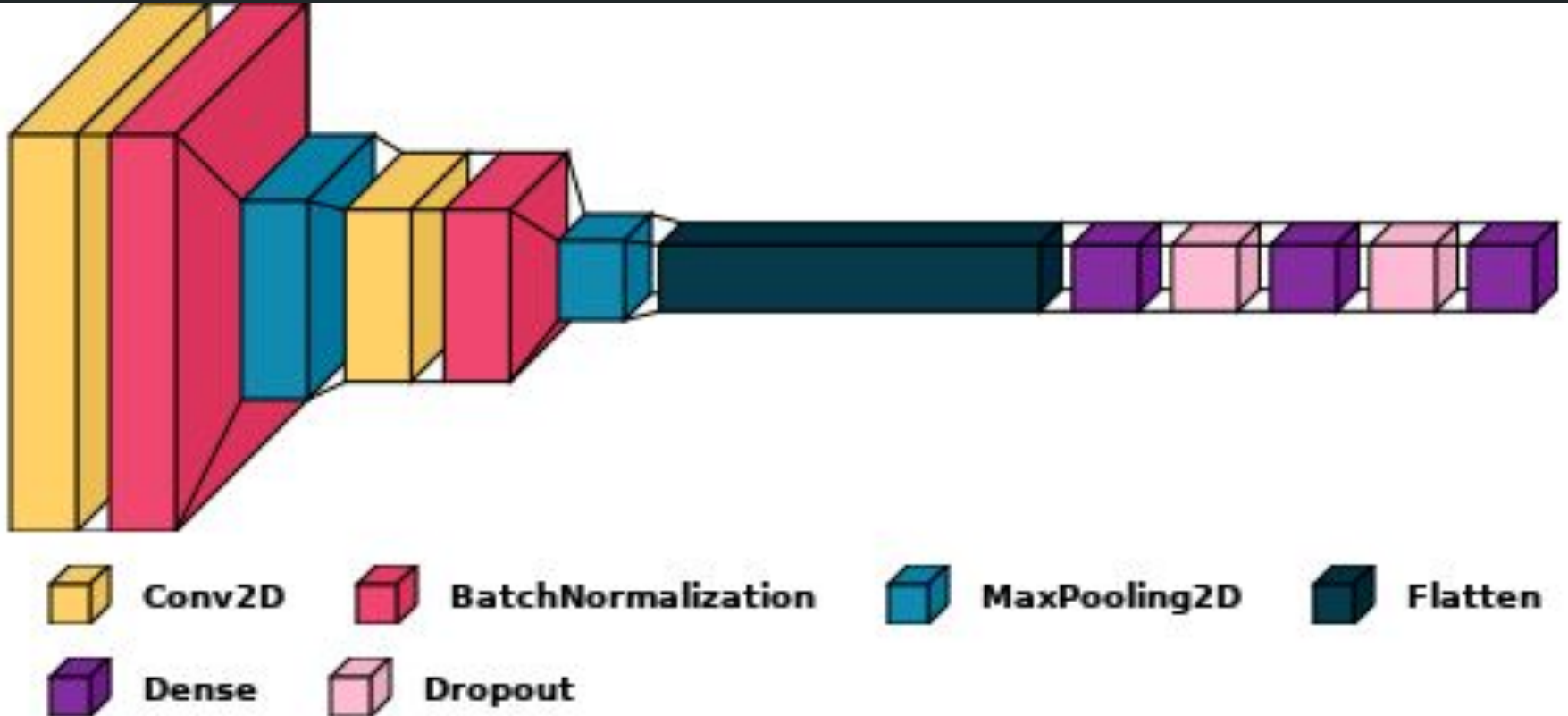


Loading Data

Preprocessing Data

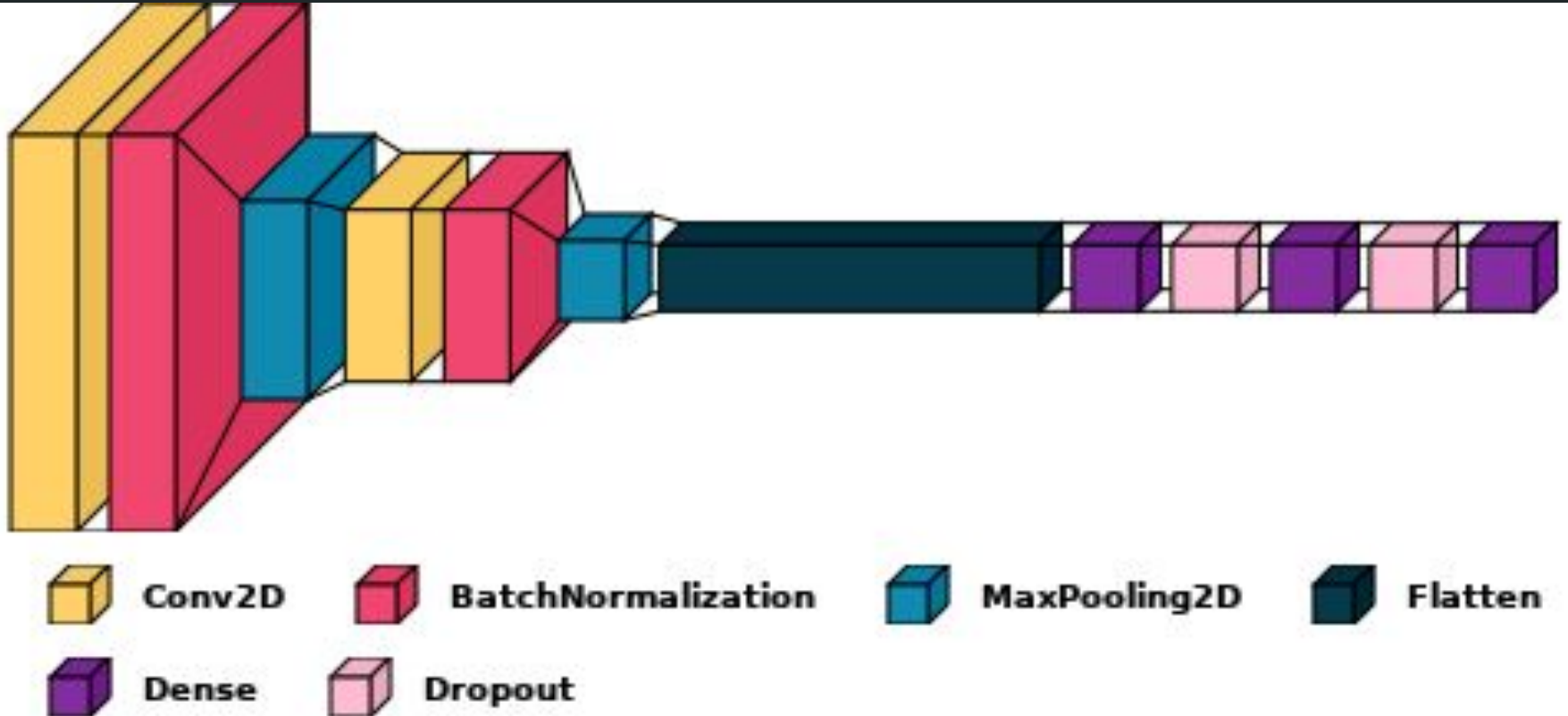
Convolutional Neural Network Architecture

DIGITS MODEL



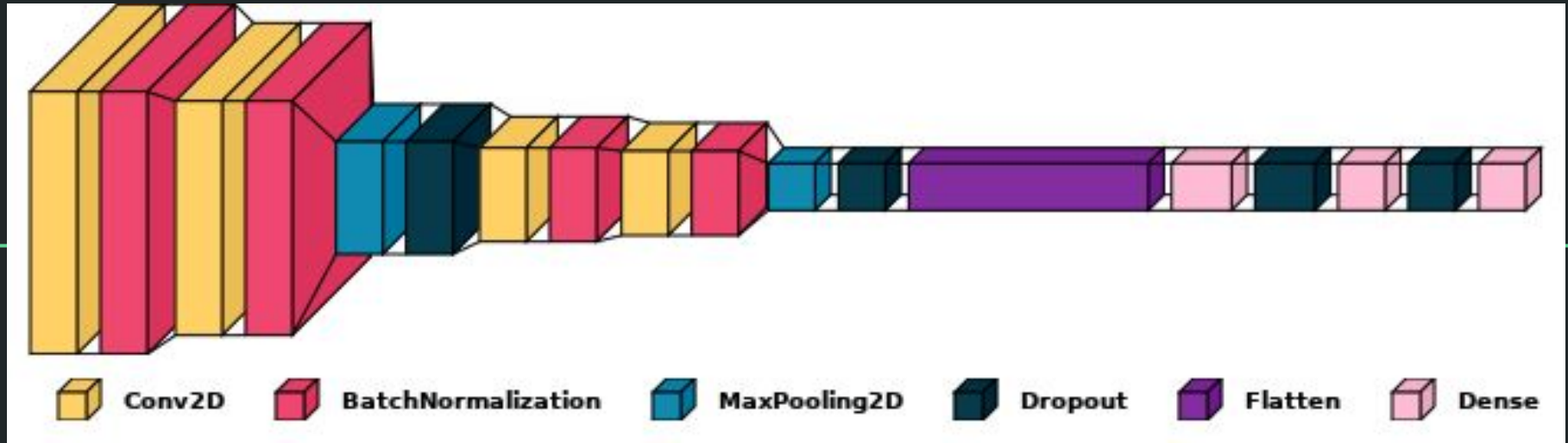
Convolutional Neural Network Architecture

LETTERS MODEL



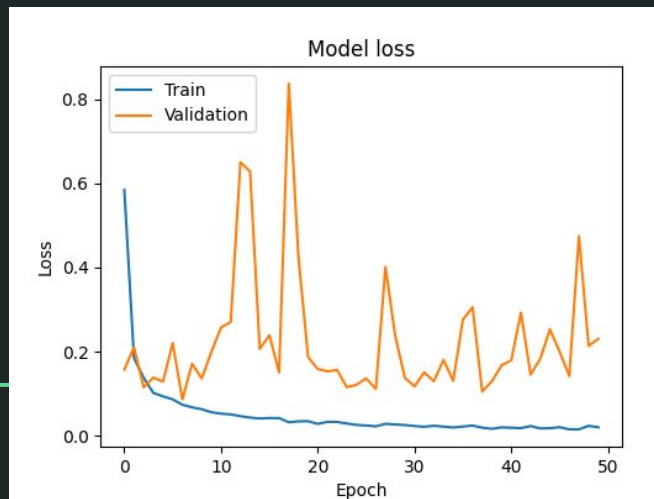
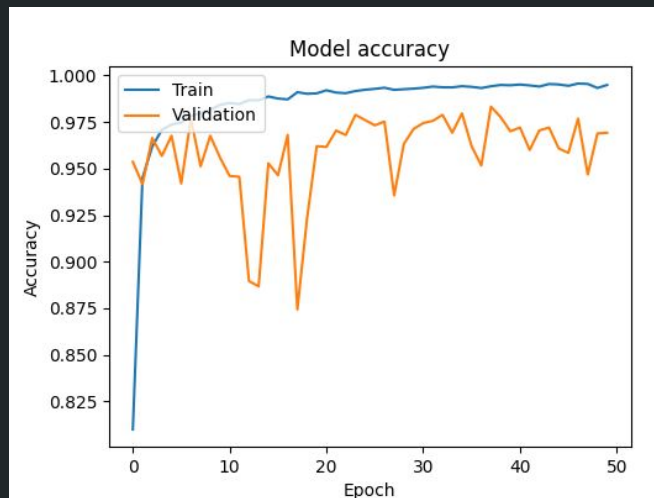
Convolutional Neural Network Architecture

CLASSIFIER MODEL



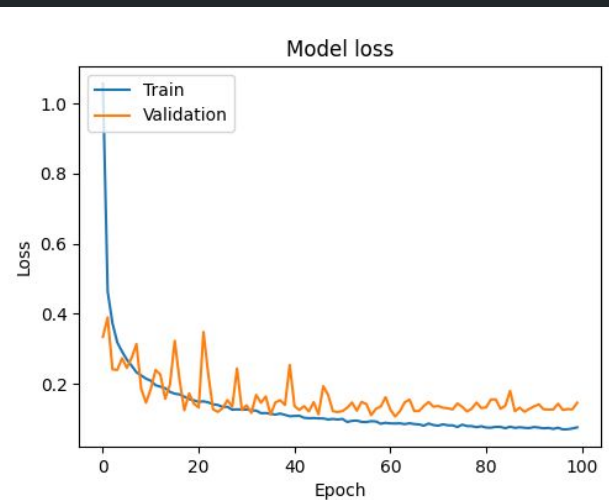
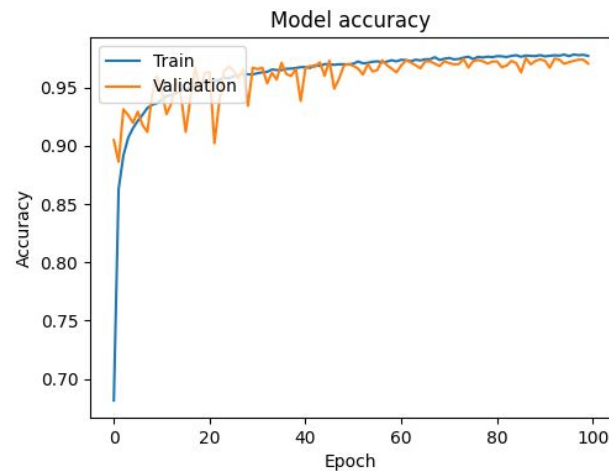
DIGIT MODEL

Accuracy & Loss



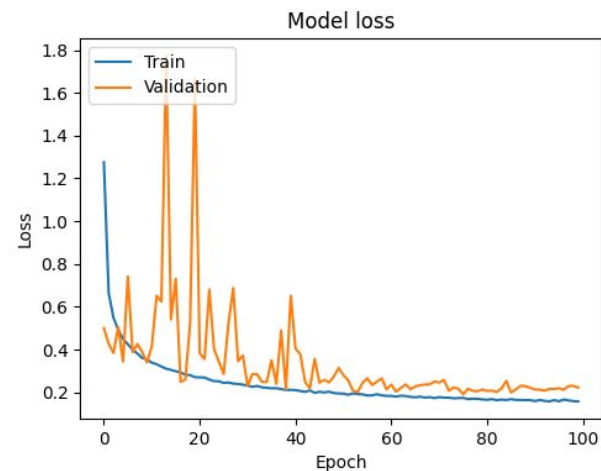
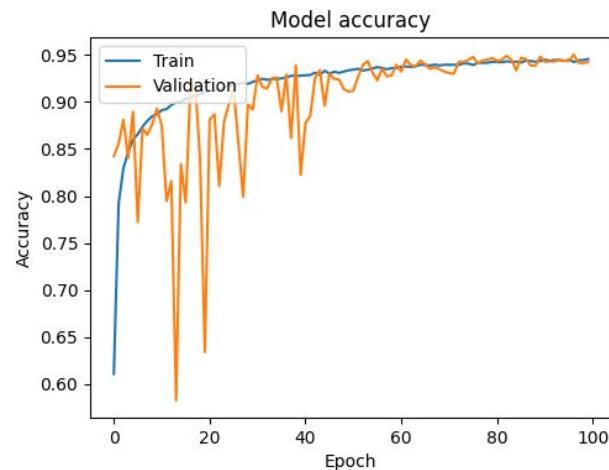
UPPERCASE MODEL

Accuracy & Loss



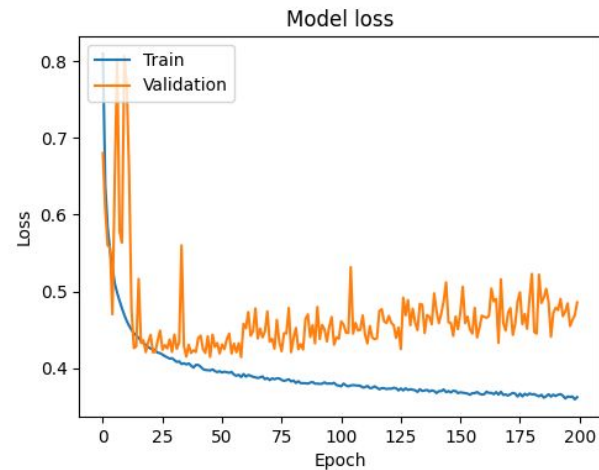
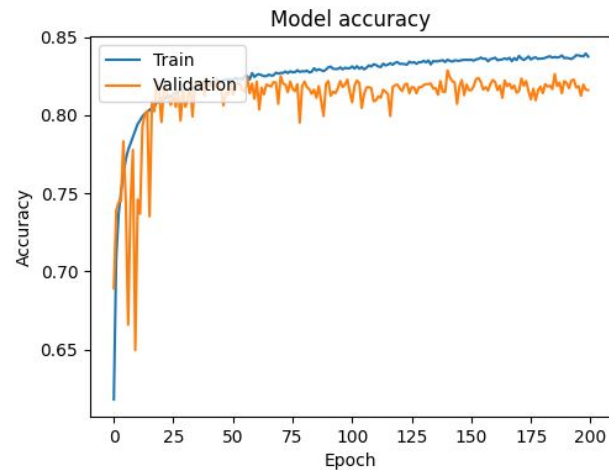
LOWERCASE MODEL

Accuracy & Loss



CLASSIFIER MODEL

Accuracy & Loss



Applications of HCR

Thank You
